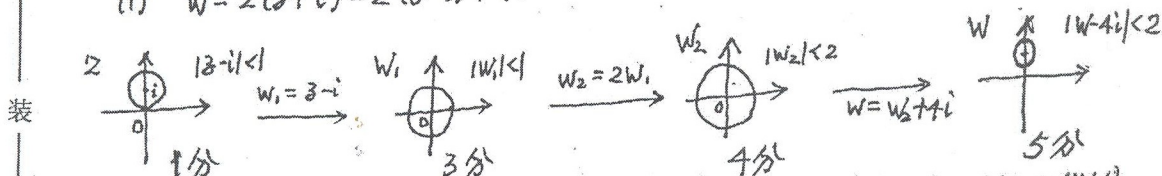


参考答案

一. (10分) 每小题5分.

(1) $W = 2(3+i) = 2(3-i) + 4i$



或者 令 $z = x+yi$, 则 $W = 2(x+yi+1) = u+vi$, 从而 $u = 2x$, $v = 2(y+1)$, 1分 3分

即 $x = \frac{u}{2}$, $y = \frac{v}{2} - 1$. 故 $|\frac{u}{2} + (\frac{v}{2} - 1)i - i| < 1$, 即 $|u+vi-4i| < 2$. 4分 5分

(2) 设 $z = x+yi$, 则 $f(z) = x^2+y^2 - i(x^2-y^2)$, 从而 $u(x,y) = x^2+y^2$, $v(x,y) = y^2-x^2$. 1分

易知 $\frac{\partial u}{\partial x} = 2x$, $\frac{\partial u}{\partial y} = 2y$, $\frac{\partial v}{\partial x} = -2x$, $\frac{\partial v}{\partial y} = 2y$, 且 u, v 在整个复平面上可微. 3分

但仅在 $x=y$ 上满足 C-R 条件, 因此 $f(z)$ 在直线 $y=x$ 上可导, 无解析点. 4分 5分

二. (6分). 解 $\frac{\partial u}{\partial x} = -\frac{2xy}{(x^2+y^2)^2}$, $\frac{\partial u}{\partial y} = \frac{x^2-y^2}{(x^2+y^2)^2}$, 故 $f'(z) = \frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}$ 2分

$= -\frac{2xy}{(x^2+y^2)^2} + \frac{y^2-x^2}{(x^2+y^2)^2}i = -\frac{1}{z^2}i$ 4分 从而 $f(z) = \frac{i}{z} + C$ (其中 $C = C_1 + C_2i$ 为复常数)

又由 $f(1) = 0$ 知 $C + i = 0$, 故 $C = -i$, 即 $f(z) = \frac{i}{z} - i = \frac{y}{x^2+y^2} + (\frac{x}{x^2+y^2} - 1)i$. 6分

三 (60分) 每小题6分.

(1) 1) z 的参数方程 $z(t) = (1+i)t$, $t \in [0, 1]$. 即 $x(t) = t$, $y(t) = t$, $dz = (1+i)dt$, 3

$\int_C (x-y+i x^2) dz = \int_0^1 t^2 (1+i) dt = \frac{1+i}{3}$ 5

2) 0) i 的参数方程为 $z(t) = ti$, $t \in [0, 1]$, 即 $x(t) = 0$, $y(t) = t$, $dz = i dt$, 1

(1) $1+i$ 的参数方程为 $z(t) = t+ti$, $t \in [0, 1]$, 即 $x(t) = t$, $y(t) = t$, $dz = dt$, 2

$\int_C (x-y+i x^2) dz = \int_0^1 (-t) i dt + \int_0^1 (t-1+it^2) dt = -\frac{1}{2} - \frac{i}{6}$. 3 4 6

(2) 由于函数 $\frac{1}{(z^2+2z+2)(z-2)}$ 在区域 $|z| \leq \frac{1}{2}$ 上解析, 故由 Cauchy 积分定理

有 $\int_{|z|=\frac{1}{2}} \frac{1}{(z^2+2z+2)(z-2)} dz = 0$. 6

自觉遵守考场纪律, 诚信考试, 绝不作弊

装

订

线

$$(3) \int_{|z|=1} \frac{\cos z}{(z^3-1)^2} dz = \int_{|z|=1} \frac{\cos z}{(z^3-1)(z^3+1)^2} dz = \int_{|z|=1} \frac{\cos z}{(z^3-1)^2} dz = \frac{2\pi i}{1!} \left[\frac{\cos z}{(z^3+1)^2} \right]' \Big|_{z=1} = -\frac{2\pi i}{9} (\sin 1 + 2\cos 1).$$

$$(4) \int_{|z|=1} \frac{dz}{(z^2+1)(z^2+2)} = \int_{|z|=1} \frac{\frac{1}{2(z^2+2)}}{z+\frac{1}{2}} dz = 2\pi i \cdot \frac{1}{2(z^2+2)} \Big|_{z=-\frac{1}{2}} = \frac{4}{9}\pi i.$$

$$(5) \int_{|z|=2} \frac{|dz|}{|z-1|^2} = \int_{|z|=2} \frac{|dz|}{(z-1)(\bar{z}-1)} = \int_{|z|=2} \frac{|dz|}{z\bar{z}-z-\bar{z}+1} = \int_{|z|=2} \frac{|dz|}{5-z-\frac{4}{z}} = \int_{|z|=2} \frac{1}{5-z-\frac{4}{z}} \cdot \frac{2}{i} dz =$$

$$= 2i \int_{|z|=2} \frac{dz}{z^2-5z+4} = 2i \cdot \int_{|z|=2} \frac{1}{z-4} dz = -4\pi \cdot \frac{1}{z-4} \Big|_{z=1} = \frac{4}{3}\pi.$$

$$(6) \int_{|z|=2} \frac{z^2}{(z^3+1)^2} dz = 2\pi i \cdot \left(\text{Res} \left[\frac{z^2}{(z^3+1)^2}, 1 \right] + \text{Res} \left[\frac{z^2}{(z^3+1)^2}, -1 \right] \right) = 2\pi i \left(\frac{1}{3} - \frac{1}{3} \right) = 0.$$

$$(7) \int_0^{2\pi} \frac{1+2\cos \theta}{5+4\cos \theta} d\theta = \int_0^{2\pi} \frac{1+2 \cdot \frac{e^{i\theta}+e^{-i\theta}}{2}}{5+4 \cdot \frac{e^{i\theta}+e^{-i\theta}}{2}} \frac{de^{i\theta}}{ie^{i\theta}} = \int_{|z|=1} \frac{1+z+z^{-1}}{5+2(z+z^{-1})} \frac{dz}{iz} = \int_{|z|=1} \frac{z^2+z+1}{iz(z^2+1)(z+2)} dz$$

$$= \frac{2\pi i}{i} \left(\text{Res} \left[\frac{z^2+z+1}{z(z^2+1)(z+2)}, 0 \right] + \text{Res} \left[\frac{z^2+z+1}{z(z^2+1)(z+2)}, -\frac{1}{2} \right] \right) = 2\pi \left(\frac{1}{2} - \frac{1}{2} \right) = 0.$$

$$(8) \int_{|z|=2} \frac{z^3}{1+z} e^{\frac{1}{z}} dz = \int_{|z|=2} \frac{z^3}{1+z} e^{\frac{1}{z}} \frac{1}{z^2} dz = \int_{|z|=2} \frac{e^{\frac{1}{z}}}{z(1+z)} dz = \frac{2\pi i}{3!} \frac{d^2}{dz^2} \left(\frac{e^{\frac{1}{z}}}{1+z} \right) \Big|_{z=0} = \frac{2\pi i}{3!} (-2) = -\frac{2\pi i}{3}.$$

(9) $\ln(1+\frac{1}{z})$ 的奇点构成了实轴上的区间 $[-1, 0]$, 因此它在区域 $1 < |z| < \infty$ 内解析,

$$\text{故 } z^2 \ln(1+\frac{1}{z}) = z^2 \cdot \sum_{n=0}^{+\infty} \frac{(-1)^n}{n+1} \left(\frac{1}{z} \right)^{n+1} = \sum_{n=0}^{+\infty} \frac{(-1)^n}{n+1} \cdot \frac{1}{z^{n-1}} \quad (z > 1, |z| < \infty).$$

$$\text{从而 } \int_{|z|=2} z^2 \ln(1+\frac{1}{z}) dz = \int_{|z|=2} \frac{\sum_{n=0}^{+\infty} \frac{(-1)^n}{n+1} \cdot \frac{1}{z^{n-1}}}{3} dz = 2\pi i \cdot \frac{(-1)^2}{3} = \frac{2\pi i}{3}.$$

⑩ (6分) $f(z) = \frac{z+1}{(z-3)(z-2)} = \frac{4}{z-3} - \frac{3}{z-2}$

$f(z)$ 在 $|z| < 2$ 内 Taylor 展开,

$$\frac{1}{z-3} = -\frac{1}{3} \cdot \frac{1}{1-\frac{z}{3}} = -\frac{1}{3} \sum_{k=0}^{+\infty} \frac{z^k}{3^k} \quad \frac{1}{z-2} = -\frac{1}{2} \cdot \frac{1}{1-\frac{z}{2}} = -\frac{1}{2} \sum_{k=0}^{+\infty} \frac{z^k}{2^k}.$$

故 $f(z) = \sum_{k=0}^{+\infty} \left(-\frac{4}{3^{k+1}} + \frac{3}{2^{k+1}} \right) z^k$. (6分).

五. (12分).

$$1) \frac{1}{(z+1)^5} = \frac{1}{24} \left(\frac{1}{z+1} \right)^{(5)}, \text{ 而 } \text{当 } |z| < +\infty \text{ 时, } \frac{1}{z+1} = \frac{1}{z} \frac{1}{1+\frac{1}{z}} = \frac{1}{z} \sum_{k=0}^{+\infty} (-1)^k \frac{1}{z^k}$$

$$= \sum_{k=0}^{+\infty} (-1)^k z^{-(k+1)}$$

$$\text{因此 } \frac{1}{(z+1)^5} = \frac{1}{24} \left(\sum_{k=0}^{+\infty} (-1)^k z^{-(k+1)} \right)^{(4)} = \frac{1}{24} \sum_{k=0}^{+\infty} (-1)^k (k+4)(k+3)(k+2)(k+1) z^{-k-5}$$

$$\text{从而 } f(z) = \sum_{k=0}^{+\infty} \frac{(-1)^k (k+4)(k+3)(k+2)(k+1)}{24} z^{-k-6}, \text{ 当 } |z| < +\infty \text{ 时.}$$

$$= \sum_{n=-\infty}^{-6} \frac{(-1)^n (n+2)(n+3)(n+4)(n+5)}{24} z^n. \quad 6$$

$$2) \text{ 当 } |z+1| < +\infty \text{ 时, } \frac{1}{z} = \frac{1}{z+1-1} = \frac{1}{z+1} \frac{1}{1-\frac{1}{z+1}} = \sum_{k=0}^{+\infty} \frac{1}{(z+1)^{k+1}} \quad 4$$

$$\text{从而 } f(z) = \frac{1}{z} \cdot (z+1)^{-5} = \sum_{k=0}^{+\infty} \frac{1}{(z+1)^{k+6}}$$

$$= \sum_{n=6}^{+\infty} (z+1)^{-n} \quad 6$$

六. (6分)

$$\text{证: 由 Cauchy 定理求导公式知 } f'(z) = \frac{1}{2\pi i} \int_{|\xi|=1} \frac{f(\xi)}{(\xi-z)^2} d\xi. \quad (2分)$$

$$\text{从而 } |f'(z)| \leq \frac{1}{2\pi} \max_{|\xi|=1} |f(\xi)| \cdot \int_{|\xi|=1} \frac{|d\xi|}{|\xi-z|^2}. \quad (4分)$$

$$\text{另外, 当 } |z|=1 \text{ 时, } |\xi-z| \geq \frac{1}{2}. \text{ 因此 } |f'(z)| \leq \frac{1}{2\pi} \cdot \int_{|\xi|=1} \frac{d\xi}{\frac{1}{4}} = 4. \quad (6分).$$